Mathematical Experiments

General Notes

Mathematics Experiments

Independent mathematical ideas,
sketches, and extended notes.

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Contents

Chapter 1

Template Independent Note

1.1 Seed Idea

Write the core idea here. This file is a placeholder for an independent mathematical note created from a discussion, a loose thought, or a larger experiment.

1.2 Working Notes

Use this section to develop definitions, examples, calculations, diagrams, and open questions. The goal is not to force the idea into a course sequence, but to make the thought readable enough that it can later be revised, expanded, or moved into a formal lecture-note project.

Remark

This chapter is intentionally independent. Future notes should usually be added as new `chapter_*.tex` files rather than inserted into this template.

1.3 Possible Next Steps

- Identify the main definitions or objects involved.
- Add one or two motivating examples.
- Record any useful diagrams in `figures/independent_notes/`.
- Decide later whether the note belongs in a formal course project.

Chapter 2

Peano Axioms and Arithmetic from Successor

2.1 Purpose of the Note

This independent note explains how the natural numbers can be described without assuming ordinary arithmetic in advance. The starting point is not a list of calculation rules, but a small structural idea: there is a first number, called 0, and there is a successor operation which produces the next number.

The main goal is to show how addition and multiplication can be introduced from this successor structure, and then to verify concrete equalities such as $2 + 3 = 5$ and $2 \cdot 2 = 4$ by unfolding the definitions.

2.2 The Peano Structure

Definition

A *Peano system* consists of a set $\mathbb{N}$, a distinguished element $0 \in \mathbb{N}$, and a function

$$S : \mathbb{N} \to \mathbb{N},$$

called the *successor function*, satisfying the following axioms.

1. 0 is a natural number.
2. If $n \in \mathbb{N}$, then $S(n) \in \mathbb{N}$.
3. 0 is not the successor of any natural number: there is no $n \in \mathbb{N}$ such that $S(n) = 0$.
4. The successor function is injective: if $S(m) = S(n)$, then $m = n$.
5. If a subset $A \subseteq \mathbb{N}$ contains 0, and whenever it contains n it also contains $S(n)$, then $A = \mathbb{N}$.

The fifth axiom is the *principle of induction*. It says that a property which holds for 0 and is preserved by the successor operation must hold for every natural number.

The usual numerals are then abbreviations:

$$1 = S(0), \qquad 2 = S(1), \qquad 3 = S(2), \qquad 4 = S(3), \qquad 5 = S(4).$$

Thus the statement $2 + 3 = 5$ should not be read as something already known. In this note it will be derived from the recursive definition of addition.

2.3 The Recursion Principle

The Peano axioms support a basic construction principle: to define a function on $\mathbb{N}$, it is enough to specify its value at 0 and explain how to pass from the value at n to the value at $S(n)$.

Theorem

Let X be a set, let $x_0 \in X$, and let $F : X \to X$ be a function. There is a unique function $f : \mathbb{N} \to X$ such that

$$f(0) = x_0, \qquad f(S(n)) = F(f(n)) \quad \text{for every } n \in \mathbb{N}.$$

In this note, the recursion principle is used to define addition first, and multiplication second. The definitions are recursive in the second argument.

2.4 Addition

Definition

For natural numbers $a, b \in \mathbb{N}$, addition is defined recursively by

$$a + 0 = a,$$

and

$$a + S(b) = S(a + b).$$

The first rule says that adding zero on the right changes nothing. The second rule says that adding the successor of b is the same as taking one successor after adding b.

Proposition 2.1. *For every $b \in \mathbb{N}$, one has*

$$0 + b = b.$$

Proof. We prove the statement by induction on b.

For $b = 0$, the definition of addition gives

$$0 + 0 = 0.$$

Assume now that $0 + b = b$. Then

$$0 + S(b) = S(0 + b) = S(b),$$

where the first equality is the recursive definition of addition and the second uses the induction hypothesis. Hence the property is preserved by successor. By induction, $0 + b = b$ for every $b \in \mathbb{N}$. □

A Formal Computation of $2 + 3$

We first compute small additions directly from the recursive rules.

Since $1 = S(0)$,

$$2 + 1 = 2 + S(0) = S(2 + 0) = S(2) = 3.$$

Since $2 = S(1)$,

$$2 + 2 = 2 + S(1) = S(2 + 1) = S(3) = 4.$$

Since $3 = S(2)$,

$$2 + 3 = 2 + S(2) = S(2 + 2) = S(4) = 5.$$

Therefore

$$\boxed{2 + 3 = 5}.$$

Notice that the computation did not assume a prior addition table. It used only the definitions of the numerals and the two recursive clauses defining addition.

2.5 Multiplication

Definition

For natural numbers $a, b \in \mathbb{N}$, multiplication is defined recursively by

$$a \cdot 0 = 0,$$

and

$$a \cdot S(b) = a \cdot b + a.$$

The definition says that multiplying by zero gives zero, and multiplying by the successor of b means multiplying by b and then adding one more copy of a.

A Formal Computation of $2 \cdot 2$

Since $1 = S(0)$,

$$2 \cdot 1 = 2 \cdot S(0) = 2 \cdot 0 + 2 = 0 + 2 = 2,$$

where the last equality follows from the proposition $0 + b = b$.

Since $2 = S(1)$,

$$2 \cdot 2 = 2 \cdot S(1) = 2 \cdot 1 + 2 = 2 + 2 = 4.$$

Therefore

$$\boxed{2 \cdot 2 = 4}.$$

Again, the equality is not taken from a multiplication table. It follows from the recursive definition of multiplication together with the previously defined addition.

2.6 What Has Been Constructed

The order of construction matters.

1. The Peano axioms give the natural numbers as a successor structure.
2. The numerals $1, 2, 3, 4, 5, \ldots$ are names for repeated successors of 0.
3. Addition is defined by recursion from the successor operation.
4. Multiplication is defined by recursion from addition.
5. Concrete arithmetic facts are then consequences of these definitions.

Summary

The Peano axioms do not start with arithmetic tables. They start with 0, the successor function S, injectivity of successor, the fact that 0 has no predecessor, and induction. From this structure one defines addition, then multiplication, and only afterward proves identities such as $2 + 3 = 5$ and $2 \cdot 2 = 4$.